

PABLO WILLIAMS

London, UK | pablowilliams119@gmail.com | +44 7508 522776 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

WORK EXPERIENCE

Applied AI Engineering Intern, QuantiFire Ltd - London, UK

May 2026 - Present

- Built an IRIS add-on agent replicating an institutional investor-relations agent, fielding questions on company filings, surveys and shareholder data through a learned routing policy across a hybrid retrieval stack spanning dense vector, BM25 keyword, schema-grounded text-to-SQL and knowledge-graph traversal, fused with reciprocal-rank fusion and HyDE query expansion.
- Engineered a deterministic-evidence pipeline in which every quantitative claim is computed as a SQL aggregate over the full response population and the language model is confined to narration, while a numeric-faithfulness verifier reconciles each figure against source and abstains on any ungrounded value, structurally precluding hallucinated figures.
- Shipped around 9,000 lines of Python with more than 700 tests passing.

Data Analyst, APJ Kapital - London, UK

Sep 2023 - May 2026

- Turn company filings, earnings calls and live market news into clear buy, hold and sell calls across a portfolio of more than 30 equities, and write up the reasoning behind each call for the weekly review.
- Build and maintain the PostgreSQL pipelines and Tableau dashboards the desk leans on each week, tracking P&L, sector allocation and factor exposure so the team can spot where risk is building before it starts to matter.
- Set up an n8n workflow that pulls the daily market-update videos off Discord, transcribes them with Whisper and sends GPT-4o summaries out over Telegram and email, sparing everyone from sitting through every clip.

AI Investment Programming Intern, MDOTM - Milan, Italy

May 2022 - Oct 2022

- Built investment tools and customer-segmentation models in Python and R, handling the preprocessing, feature engineering and supervised learning end to end so they predicted and targeted more accurately.
- Worked in a 40-person team to fold ML into the day-to-day investment workflow, running data-quality reviews and turning what stakeholders flagged into measurable performance gains.
- Helped build MDOTM's ML platform, which went on to win the 2022 AIFIN Financial Innovation Award and Best Product of the Year at the New York FinTech Awards.

PROJECTS

ToneShift | React, TypeScript, Supabase, FinBERT

Mar 2026

FinBERT fine-tuned on 1.1M SEC filings to score tone sentence by sentence, roll it up across a document and pick out hedging. Hits 0.81 F1 at 340ms and catches tone shifts before analyst consensus moves.

Optimal Execution Platform | Python, PyTorch, PPO, LOBSTER

Apr 2026

Pits three execution strategies against live LOBSTER AAPL order-book data: the Almgren-Chriss closed form, Obizhaeva-Wang transient impact, and a PPO agent trained on a simulated book.

Calibration-Optimised Equity Outperformance | Python, XGBoost, LightGBM, Optuna, SHAP

Jan 2026

Asked whether better-calibrated classifiers actually trade better, sizing positions half-Kelly across 457 S&P 500 stocks from 2010 to 2024. Random Forest calibrated best (ECE 0.026) and returned 68.1% at a 0.77 Sharpe.

Neural Volatility Surface Arbitrage Scanner | Python, PyTorch, Three.js, WebGL

Apr 2026

Fits an SVI volatility surface with a neural residual head to tidy the fit, checks it for calendar and butterfly arbitrage, and renders the result as an interactive 3D dashboard.

RiskTerrain | Python, LangGraph, SurrealDB, Docker

Mar 2026

A 6-node LangGraph pipeline that reads live geopolitical events, walks supply-chain links across 154 S&P 500 companies, and scores the knock-on risk in under 8 seconds. Built at the LangChain x SurrealDB London Hackathon.

Attention Lens | JavaScript, Transformers.js, WebGPU, GPT-2

Apr 2026

Runs a 124M-parameter GPT-2 entirely in the browser through Transformers.js and WebGPU, so you can watch attention as it happens across a heatmap, a connection view and all 144 heads in 12 layers.

Grokking Circuits | PyTorch, Transformers, Fourier

Dec 2024

Pulls apart a 1-layer transformer trained on addition mod 113 to see how it learns, recovering the Fourier circuit it leans on, where six frequencies alone hold 94.5% test accuracy, with grokking kicking in at step 10,900.

Interactive ML Monographs & Monte Carlo Terminals | JavaScript, Python, WebGPU, ONNX

2024 - 2026

19 interactive ML monographs (ray tracing, optimal transport, Kalman filtering, RAG, CPython internals) and 10 Monte Carlo terminal dashboards across finance, physics and biodiversity, all written between 2024 and 2026 and running in the browser (WCAG 2.2 AA).

EDUCATION

MSc Business Analytics, UCL

Sep 2025 - Present

Predicted: Distinction (72%); focus on artificial intelligence and natural language processing. Modules: Natural Language Processing, Machine Learning for Domain Specialists, Predictive Analytics, Programming for Analytics, Statistical Foundations, Mathematical Analysis, Data Visualisation, Marketing Analytics, Operations Analytics, and Business Strategy.

BSc Business Analytics (Minor: Accounting & Finance), Lancaster University

Sep 2021 - Jul 2025

Achieved: Upper Second-Class Honours (2:1), with an Upper Second in the Accounting & Finance minor. Modules: Data Mining, Data Visualisation, Business Statistics, Operations Management, Database Systems, Simulation Modelling, Decision Analysis, Spreadsheet Modelling, and Statistical Analytics.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, DuckDB), R (tidyverse, ggplot2), JavaScript

ML: XGBoost, LightGBM, Random Forest, PyTorch, TensorFlow, LLMs (LangChain, LangGraph, Ollama), Optuna, SHAP

Data Engineering: PySpark, MongoDB, DuckDB, Parquet, Docker, n8n, AWS (S3), GitHub Actions

Visualisation & BI: Tableau, Streamlit, Matplotlib, Seaborn, LaTeX | **NLP & LLM Ops:** Sentence-Transformers, RAG, GraphRAG, Guardrails, Whisper, GPT-4o

CERTIFICATIONS

Bloomberg: Finance Fundamentals, Market Concepts, Spreadsheet Analysis | **Google:** Data Analytics, IT Automation with Python | **IBM:** BI Analyst, Applied Software Engineering, AI Enterprise Workflow | **Yale:** Financial Markets | **LBS:** Financial Analysis | **Duke:** MLOps

LEADERSHIP, LANGUAGES & INTERESTS

Mathematics & CS Tutor - GCSE, IB, SAT, GMAT levels; tracking student progress with Excel and Tableau diagnostics.

Feb 2024 - Present

Data Science Society Member - Assisted peers with ML and analytics projects weekly.

Oct 2022 - May 2025

Volunteer, Milan Dog Kennel - Walking and caring for rescue dogs every Sunday.

Sep 2015 - Sep 2021

Languages: Italian (Native) | English (Native) | Spanish (Fluent) | German (Beginner)